Daniel Domínguez Trejo

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Summary

Data Engineering graduate (UPM) who builds tested, documented data pipelines. Delivered an end-to-end computer-vision solution at ArcelorMittal Global R&D (**82% IoU**; Bachelor's thesis graded **10/10**) and a daily ELT pipeline on **Airflow, dbt and DuckDB**. Looking for a junior data engineering role in Europe.

Experience

ArcelorMittal Global R&D South (in collaboration with OCAS)
Research Engineering Apprentice Mar – Jul 2025 · Oct – Dec 2025

- Built a **U-Net semantic segmentation model in PyTorch** that classifies every pixel of steel micrographs as steel, resin or copper-infiltration defect, reaching **82% IoU** against manual labels.
- Labelled the training images in CVAT and built the preprocessing and evaluation workflow for microscopy data from different samples and lab sessions.
- Characterised each detected defect with scikit-image (area, orientation, aspect ratio) and measured surface roughness at the steel–resin interface, turning a manual inspection into an analysis that takes **seconds**.
- Shipped a **Gradio web app** so metallurgical technicians, with no programming background, can inspect predictions and download the results.

Projects

Madrid Air Quality & Weather Pipeline Live dashboard · GitHub · 2026

- Daily **ELT pipeline** joining hourly data from Madrid's 24 air-quality stations with Open-Meteo weather (~3M measurements since 2024), orchestrated with **Airflow** (Cosmos) in Docker Compose.
- **dbt models on DuckDB** handle unpivoting, UTC and daylight-saving conversion and source precedence; results published as a Quarto dashboard on GitHub Pages.
- **72 automated tests** (pytest, dbt data and unit tests) run in GitHub Actions CI on every push; 34 design decisions documented in the repository.

Education

Universidad Politécnica de Madrid — B.Sc. in Data Engineering 2026

Bachelor's thesis (**10/10**): *Application of Computer Vision techniques to streamline and automate the characterization of steel micrographs*, developed with ArcelorMittal Global R&D.

Skills & Languages

Programming: Python, SQL, C, JavaScript

Data engineering: Airflow, dbt, DuckDB, Docker, Git, GitHub Actions, pytest, uv

ML / computer vision: PyTorch, U-Net, scikit-image, CVAT, Gradio, Jupyter

Languages: Spanish (native) · English (C1) · French (A2)

Additional

Part-time work in a family-owned bookstore alongside university studies: customer service and day-to-day operations.